

A Fuzzy Nesterov-Accelerated Nonnegative Latent Factorization of Tensors Model for Efficient Representation of Dynamic Directed Graph Supplementary File

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I. INTRODUCTION

THIS is the supplementary file for the paper entitled “A Fuzzy Nesterov-Accelerated Nonnegative Latent Factorization of tensors Model for Efficient Representation to Dynamic Directed Graph”. We provide the proofs of Lemmas 1–2 and Theorems 1–2 in Section II, the FNL algorithm in Section III, and the tables and figures of the experimental results in Section IV.

II. MODEL CONVERGENCE ANALYSIS

The goal of the FNL model is to represent an HDI tensor $\mathbf{Y}$ via a nonnegative latent factorization. The optimization problem is defined as:

$$\min_{\mathbf{U}, \mathbf{W}, \mathbf{Z}} \varepsilon(\mathbf{U}, \mathbf{W}, \mathbf{Z}) = \sum_{(i,j,k) \in \Lambda} (y_{ijk} - \hat{y}_{ijk})^2 + \lambda^t (\sigma(u_{ir})^2 + \sigma(w_{jr})^2 + \sigma(z_{kr})^2), \quad (\text{S1})$$

where $\hat{y}_{ijk} = \sum_{r=1}^R \sigma(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr})$, Λ and λ_t represent the known-element set and the regularization coefficient at the t -th iteration, respectively, and $\sigma(\cdot)$ represents the hard-sigmoid function:

$$\sigma(\theta) = \max(0, \min(1, 0.2\theta + 0.5)), \quad (\text{S2})$$

where Λ and λ_t represent the known-element set and the regularization coefficient at the t -th iteration, respectively.

For convenience in the analysis, we alternately fix the counterparts of the active parameter as constants, i.e., we treat w_{jr} and z_{kr} as constant when analyzing u_{ir} . The gradient with respect to u_{ir} is:

$$\nabla_{u_{ir}} \varepsilon_{i,j,k} = - \left(y_{ijk} - \sum_{r=1}^R \sigma(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr}) \right) \sigma'(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr}) + \lambda^t u_{ir}. \quad (\text{S3})$$

Note that the NASGD update of u_{ir} at the t -th iteration is given as follows.

$$\begin{aligned} t = 1 : u_{ir}^{t+1} &= u_{ir}^t - \eta^t (1 + \gamma^t) \nabla_{u_{ir}} \varepsilon_{i,j,k} (u_{ir}^t) \\ t \geq 2 : u_{ir}^{t+1} &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{i,j,k} (u_{ir}^t) \\ &\quad + \gamma^t ((u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{i,j,k} (u_{ir}^t)) - (u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{i,j,k} (u_{ir}^{t-1}))). \end{aligned} \quad (\text{S4})$$

According to the fuzzy reasoning rule table (Table I), we obtain:

$$\begin{aligned} \eta^t &\in [\eta_{min}, \eta_{max}], 0 < \eta_{min} \leq \eta^t \leq \eta_{max}, \\ \lambda^t &\in [\lambda_{min}, \lambda_{max}], 0 < \lambda_{min} \leq \lambda^t \leq \lambda_{max}, \\ \gamma^t &\in [\gamma_{min}, \gamma_{max}], 0 < \gamma_{min} \leq \gamma^t \leq \gamma_{max} < 1. \end{aligned} \quad (\text{S5})$$

In addition, to ensure the stability of the NASGD iterations and an effective error bound, we assume that the gradient satisfies Lipschitz continuity with constant L^t .

Assumption 1: Since the function is convex, the gradient satisfies

$$\|\nabla_{u_{ir}} \varepsilon_{i,j,k}^t(u) - \nabla_{u_{ir}} \varepsilon_{i,j,k}^t(v)\| \leq L^t |u - v|, \quad (\text{S6})$$

where u and v represent two different points in the domain. Next, we calculate:

$$\begin{aligned} \nabla_{u_{ir}} \varepsilon_{i,j,k}^t(u) - \nabla_{u_{ir}} \varepsilon_{i,j,k}^t(v) &= q_{jr} t_{kr} [(y_{ijk} - \hat{y}_{ijk}(v)) \sigma'(v) - (y_{ijk} - \hat{y}_{ijk}(u)) \sigma'(u)] + \lambda^t (u - v) \\ &= q_{jr} t_{kr} ((y_{ijk} - \hat{y}_{ijk}(v)) (\sigma'(v) - \sigma'(u)) + \sigma'(u) (\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v))) + \lambda^t (u - v). \end{aligned} \quad (\text{S7})$$

Since $\sigma(x) \leq 1$ and the maximum slope is 0.2, we have:

$$|\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v)| = |\sigma(u) - \sigma(v)| q_{jr} t_{kr} \leq 0.2 |u - v| q_{jr} t_{kr}, \quad |\sigma'(u) - \sigma'(v)| \leq 0.2. \quad (\text{S8})$$

Letting $Q = \max_{j,r} q_{jr} t_{kr}$, we obtain:

$$\|\nabla_{u_{ir}} \varepsilon_{i,j,k}^t(u) - \nabla_{u_{ir}} \varepsilon_{i,j,k}^t(v)\| \leq Q (|y_{ijk} - \hat{y}_{ijk}(v)| \cdot 0.2 + 0.2 + 0.2 |u - v| Q) + \lambda^t |u - v|. \quad (\text{S9})$$

Letting $M = \max |y_{ijk}|$ be a constant, we obtain:

$$|\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v)| \leq M + Q. \quad (\text{S10})$$

In summary,

$$\|\nabla_{u_{ir}} \varepsilon_{i,j,k}^t(u) - \nabla_{u_{ir}} \varepsilon_{i,j,k}^t(v)\| \leq L^t |u - v| = 0.04Q^2 + \lambda^t. \quad (\text{S11})$$

Since L^t is bounded by the bounded derivative of the hard_Sigmoid function and the regularization term, the assumption that the gradient satisfies Lipschitz continuity holds.

We define ω^t as:

$$\omega^t = \begin{cases} \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^t \nabla_{u_{ir}} \varepsilon_{i,j,k}(u_{ir}^{t-1})), & t \geq 1 \\ 0, & t = 0 \end{cases}, \quad (\text{S12})$$

Since $\eta^t, \lambda^t, \gamma^t$ are dynamically adjusted according to the fuzzy reasoning $I^{(t)} = 0.5(\text{RMSE} + \text{MAE})$, we present the following results.

Lemma 1: For any $t \geq 0$, the following equation holds:

$$\begin{aligned} u_{ir}^{t+1} + \omega^{t+1} &= u_{ir}^{t+1} + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^{t+1} - u_{ir}^t + \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \gamma^t ((u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)) - (u_{ir}^{t-1} - \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1}))) \\ &\quad + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^{t+1} - u_{ir}^t + \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \gamma^t (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &\quad + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} \gamma^t (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \left(\gamma^t + \frac{\gamma^{t+1} \gamma^t}{1-\gamma^{t+1}} \right) (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \gamma^t \left(1 + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} \right) (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \gamma^t \frac{1-\gamma^{t+1} + \gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \frac{\gamma^t}{1-\gamma^{t+1}} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \end{aligned} \quad (\text{S13})$$

$$\begin{aligned}
&= u_{ir}^t + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) - \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) - \frac{\gamma^t}{1-\gamma^t} \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \\
&= u_{ir}^t + \omega^t - \frac{\gamma^t}{1-\gamma^t} \eta^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t).
\end{aligned}$$

Lemma 2: Let $u_{ir}^{-1} = u_{ir}^0$. For any $t \geq 0$, $\exists u_{ir}^*$ such that:

$$\begin{aligned}
&\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 \\
&= \left\| u_{ir}^t + \omega^t - \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) - u_{ir}^* \right\|^2 \\
&= \left\| (u_{ir}^t + \omega^t - u_{ir}^*) - \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\|^2 \\
&= (u_{ir}^t + \omega^t - u_{ir}^*)^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle + \left(\frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right)^2 \\
&= \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 \|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\|^2.
\end{aligned} \tag{S14}$$

Note that (S13) involves u_{ir}^{-1} when $t = 0$. By setting $u_{ir}^{-1} = u_{ir}^0$, the above inequality still holds. Hence, Lemma 2 holds.

Based on Lemmas 1–2, we obtain the following result.

Theorem 1 (Convergence Bound of NASGD): For any $t \geq 0$, let $\|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\| \leq G$, $|u_{ir}^{t+1} - u_{ir}^t| \leq A$, where G and A are both positive constants. Let $B = A + 2\eta_{max}G$ be a positive constant. When $t \in \{0, \dots, T\}$, based on Lemma 2, we have:

$$\begin{aligned}
\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 &= \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle \\
&\quad + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 \|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\|^2.
\end{aligned} \tag{S15}$$

First, since the function $\varepsilon_{ijk}(u_{ir})$ is convex, for $(u_{ir}^t + \omega^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)$, we have:

$$\begin{aligned}
&(u_{ir}^t + \omega^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \\
&= (u_{ir}^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \omega^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \\
&\geq \varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*) + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t).
\end{aligned} \tag{S16}$$

Second, since y_{ijk} , $\sigma(\cdot)$, and λ_t are bounded, setting $\|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\| \leq G$, we have:

$$\begin{aligned}
&|\omega^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)| \\
&\leq \frac{\gamma^t}{1-\gamma^t} G |u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})| \\
&\leq \frac{\gamma^t}{1-\gamma^t} G \gamma^{t-1} |u_{ir}^{t-1} - u_{ir}^{t-2} + \eta^{t-2} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-2}) - \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})| \\
&\leq \frac{\gamma_{max}}{1-\gamma_{max}} G (A + 2\eta_{max}G).
\end{aligned} \tag{S17}$$

Finally, substituting (S16) and (S17) into (S15) yields:

$$\begin{aligned}
\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 &\leq \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \\
&\quad + 2 \frac{\eta^t}{1-\gamma^t} \frac{\gamma_{max}}{1-\gamma_{max}} GB + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 G^2.
\end{aligned} \tag{S18}$$

Summing from $t = 0$ to T gives:

$$\begin{aligned} \|u_{ir}^{T+1} + \omega^{T+1} - u_{ir}^*\|^2 &\leq \|u_{ir}^0 + \omega^0 - u_{ir}^*\|^2 - 2 \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\ &\quad + 2 \frac{\gamma_{\max}}{1-\gamma_{\max}} GB \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} \right) + G^2 \sum_{t=0}^T \left(\left(\frac{\eta^t}{1-\gamma^t} \right)^2 \right). \end{aligned} \quad (\text{S19})$$

Since $\omega^0 = 0$, $\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 \geq 0$:

$$\begin{aligned} &\sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\ &\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2} + \frac{\gamma_{\max}}{1-\gamma_{\max}} GB \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} \right) + \frac{1}{2} G^2 \sum_{t=0}^T \left(\left(\frac{\eta^t}{1-\gamma^t} \right)^2 \right) \\ &\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2} + \frac{\eta_{\max}}{1-\gamma_{\max}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB (T+1) + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}} (T+1). \end{aligned} \quad (\text{S20})$$

And then,

$$\begin{aligned} &\frac{1}{T+1} \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\ &\leq \frac{\sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right)}{\frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} \\ &\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2 \frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} + \frac{\eta_{\max}}{1-\gamma_{\min}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}}. \end{aligned} \quad (\text{S21})$$

Let $\bar{u}_{ir} = \sum_{t=0}^T u_{ir}^t / (T+1)$, by convexity, we obtain:

$$\begin{aligned} &\varepsilon_{ijk}(\bar{u}_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*) \\ &\leq \frac{1}{T+1} \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\ &\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2 \frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} + \frac{\eta_{\max}}{1-\gamma_{\min}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}}. \end{aligned} \quad (\text{S22})$$

The proof of Theorem 1 follows because fuzzy reasoning ensures that η_t , λ_t , and γ_t remain within bounded ranges. By the same argument, $\varepsilon_{i,j,k}(w_{jr})$ converges by training w_{jr} while fixing u_{ir} and z_{kr} as constants, and $\varepsilon_{ijk}(Z_{kr})$ converges by training z_{kr} while fixing u_{ir} and w_{jr} as constants. Therefore, Theorem 1 holds.

III. FNL ALGORITHM

Algorithm I: FNL	
Input: Λ, I, J, K, R	
Operation	Cost
/* Initialization */	
1. Initialize $P, Q, T, V_{(U)}, V_{(W)}, V_{(Z)}$ Initialize U, W, Z at random in $(0, 0.05)$ Initialize $\eta, \lambda, \gamma, R = 10, T = \text{maximum round count}$	C_1
/* Training */	
/* Parameters Update Calculation */	
2. while not converge and $t \leq T$ do	
3. for each $y_{ijk} \in \Lambda$	
4. $\hat{y}_{ijk} = 0$	
5. for $r = 1$ to R	
6. Compute p_r, q_r, t_r and $\hat{y}_{ijk}$ based on (7), (8)	
7. end for	C_2
8. $err = y_{ijk} - \hat{y}_{ijk}$	
9. for $r = 1$ to R	
10. Update $U, W, Z, V_{(U)}, V_{(W)}, V_{(Z)}$ based on (17)	
11. end for	
12. end for	
/* Hyper-parameter Self-adaptation Calculation */	
13. Computing $\Delta(t)$ and $I^{(t)}$ based on (18)	
14. Computing deg1 and deg2 based on (19)	
15. Computing $\eta^{(t)}, \lambda^{(t)}, \gamma^{(t)}$ based on (20) and table S1	
16. $t = t + 1$	
17. end while	
/* Map Matrices Calculation */	
18. for $r = 1$ to R	
19. Map U, W, Z back to P, Q, T based on (7)	C_3
20. end for	
/* Operation Ending */	
Output: P, Q, T built on Λ	

IV. EXPERIMENTAL RESULTS

TABLE S1
DETAILS OF THE FUZZY RULES

Δ	η	λ	γ
0.00001	2^{-7}	10^{-1}	0.8
0.0001	2^{-6}	10^{-2}	0.81
0.001	2^{-5}	10^{-3}	0.82
0.01	2^{-4}	10^{-4}	0.83
0.1	2^{-3}	10^{-5}	0.84
1	2^{-2}	10^{-6}	0.85

TABLE S2
HYPERPARAMETER SETTINGS OF THE COMPARED MODELS

Datasets	Hyper-parameter Setting					
D1	M2: $\lambda = 10^{-5}, \eta = 10^{-3}$ M7: $\eta = 10^{-2}, \lambda = 10^{-2}$	M3: $\lambda = 10^{-5}, \eta = 10^{-3}$ M8: $\eta = 10^{-1}, \lambda = 5 \times 10^{-3}$	M4: $\lambda = 4 \times 10^{-8}, \gamma = 480$ M9: $\eta = 10^{-1}, \lambda = 10^{-1}$	M5: $\lambda_1 = 0.04, \lambda_2 = 0.2$ M10: $\eta = 10^{-1}, \lambda = 10^{-3}$	M6: $\lambda_1 = 0.5, \lambda_2 = 0.0625$ M11: $\eta = 0.2, \lambda = 0.05$	
D2	M2: $\lambda = 10^{-5}, \eta = 10^{-3}$ M7: $\eta = 10^{-2}, \lambda = 5 \times 10^{-2}$	M3: $\lambda = 10^{-5}, \eta = 0.002$ M8: $\eta = 10^{-2}, \lambda = 10^{-3}$	M4: $\lambda = 10^{-8}, \gamma = 500$ M9: $\eta = 0.2, \lambda = 10^{-1}$	M5: $\lambda_1 = 0.04, \lambda_2 = 0.2$ M10: $\eta = 10^{-1}, \lambda = 10^{-4}$	M6: $\lambda_1 = 0.125, \lambda_2 = 0.25$ M11: $\eta = 0.2, \lambda = 10^{-2}$	
D3	M2: $\lambda = 10^{-5}, \eta = 10^{-3}$ M7: $\eta = 10^{-1}, \lambda = 10^{-1}$	M3: $\lambda = 10^{-5}, \eta = 0.004$ M8: $\eta = 10^{-1}, \lambda = 10^{-2}$	M4: $\lambda = 10^{-8}, \gamma = 450$ M9: $\eta = 10^{-1}, \lambda = 0.2$	M5: $\lambda_1 = 0.005, \lambda_2 = 0.4$ M10: $\eta = 0.2, \lambda = 10^{-2}$	M6: $\lambda_1 = 0.25, \lambda_2 = 0.0625$ M11: $\eta = 0.05, \lambda = 10^{-3}$	
D4	M2: $\lambda = 10^{-5}, \eta = 0.002$ M7: $\eta = 10^{-1}, \lambda = 5 \times 10^{-3}$	M3: $\lambda = 10^{-5}, \eta = 0.005$ M8: $\eta = 10^{-2}, \lambda = 10^{-3}$	M4: $\lambda = 4 \times 10^{-8}, \gamma = 500$ M9: $\eta = 10^{-1}, \lambda = 10^{-1}$	M5: $\lambda_1 = 0.007, \lambda_2 = 0.4$ M10: $\eta = 0.2, \lambda = 10^{-3}$	M6: $\lambda_1 = 0.5, \lambda_2 = 0.0625$ M11: $\eta = 10^{-1}, \lambda = 10^{-2}$	
D5	M2: $\lambda = 10^{-4}, \eta = 10^{-3}$ M7: $\eta = 10^{-1}, \lambda = 2 \times 10^{-2}$	M3: $\lambda = 10^{-4}, \eta = 0.008$ M8: $\eta = 10^{-1}, \lambda = 10^{-2}$	M4: $\lambda = 10^{-7}, \gamma = 480$ M9: $\eta = 0.2, \lambda = 0.2$	M5: $\lambda_1 = 0.07, \lambda_2 = 0.3$ M10: $\eta = 10^{-1}, \lambda = 0.002$	M6: $\lambda_1 = 0.5, \lambda_2 = 0.0625$ M11: $\eta = 10^{-1}, \lambda = 10^{-3}$	
D6	M2: $\lambda = 10^{-5}, \eta = 10^{-3}$ M7: $\eta = 10^{-2}, \lambda = 10^{-3}$	M3: $\lambda = 10^{-5}, \eta = 0.004$ M8: $\eta = 10^{-1}, \lambda = 5 \times 10^{-2}$	M4: $\lambda = 10^{-8}, \gamma = 450$ M9: $\eta = 10^{-1}, \lambda = 10^{-1}$	M5: $\lambda_1 = 0.006, \lambda_2 = 0.5$ M10: $\eta = 0.2, \lambda = 10^{-3}$	M6: $\lambda_1 = 0.25, \lambda_2 = 0.03125$ M11: $\eta = 10^{-1}, \lambda = 10^{-2}$	
D7	M2: $\lambda = 10^{-5}, \eta = 10^{-2}$ M7: $\eta = 10^{-2}, \lambda = 5 \times 10^{-2}$	M3: $\lambda = 10^{-5}, \eta = 0.004$ M8: $\eta = 10^{-2}, \lambda = 10^{-2}$	M4: $\lambda = 10^{-8}, \gamma = 480$ M9: $\eta = 10^{-1}, \lambda = 0.2$	M5: $\lambda_1 = 0.005, \lambda_2 = 0.4$ M10: $\eta = 0.2, \lambda = 10^{-2}$	M6: $\lambda_1 = 0.25, \lambda_2 = 0.0625$ M11: $\eta = 0.05, \lambda = 10^{-2}$	
D8	M2: $\lambda = 10^{-4}, \eta = 10^{-2}$ M7: $\eta = 10^{-1}, \lambda = 10^{-2}$	M3: $\lambda = 10^{-5}, \eta = 0.004$ M8: $\eta = 10^{-2}, \lambda = 10^{-3}$	M4: $\lambda = 10^{-8}, \gamma = 500$ M9: $\eta = 10^{-1}, \lambda = 0.2$	M5: $\lambda_1 = 0.004, \lambda_2 = 0.5$ M10: $\eta = 10^{-1}, \lambda = 10^{-2}$	M6: $\lambda_1 = 0.5, \lambda_2 = 0.03125$ M11: $\eta = 10^{-1}, \lambda = 10^{-3}$	

TABLE S3
SINGLE-TRAINING TIME TO REACH THE LOWEST RMSE/MAE (SECONDS) AND WIN/LOSS COUNTS OF M1–M11 ON D1–D8

Datasets	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11
D1	RMSE 37.48 _{±0.66} MAE 39.82 _{±0.55}	1222.25 _{±2428.2} 1318.37 _{±468.16}	1358.66 _{±61.69} 1357.62 _{±61.76}	57.01 _{±5.65} 55.97 _{±3.42}	180.48 _{±7.25} 92.55 _{±0.77}	358.29 _{±6.03} 206.14 _{±2.85}	2994.51 _{±155.71} 3322.29 _{±159.47}	4131.20 _{±252.00} 4759.32 _{±271.28}	1419.7 _{±113.98} 1812.03 _{±132.77}	522.58 _{±10.01} 1097.77 _{±54.89}	1574.86 _{±111.01} 2207.54 _{±155.04}
D2	RMSE 22.74 _{±0.35} MAE 24.17 _{±2.35}	693.79 _{±40.64} 740.65 _{±41.59}	700.9 _{±51.93} 689.27 _{±44.32}	38.83 _{±2.82} 38.83 _{±1.73}	79.12 _{±5.57} 61.89 _{±3.34}	96.65 _{±3.98} 54.67 _{±3.91}	1824.67 _{±80.29} 1836.81 _{±95.51}	2372.76 _{±116.27} 2629.31 _{±144.61}	986.4 _{±119.81} 1222.48 _{±132.77}	378.21 _{±3.17} 857.7 _{±2.31}	1214.03 _{±124.56} 1646.96 _{±211.94}
D3	RMSE 4.99 _{±0.43} MAE 3.23 _{±0.32}	270.32 _{±95.46} 272.08 _{±96.06}	122.41 _{±3.54} 119.19 _{±4.27}	31.32 _{±3.03} 30.5 _{±4.79}	25.05 _{±0.98} 15.03 _{±1.11}	61.74 _{±4.18} 51.38 _{±2.95}	732.57 _{±30.04} 642.11 _{±28.89}	989.37 _{±46.50} 889.70 _{±45.37}	394.79 _{±11.95} 482.28 _{±7.46}	175.21 _{±2.63} 342.87 _{±6.23}	604.73 _{±80.25} 783.15 _{±97.01}
D4	RMSE 8.03 _{±0.55} MAE 4.59 _{±0.66}	171.97 _{±60.97} 182.39 _{±69.95}	129.93 _{±0.28} 128.18 _{±1.32}	22.63 _{±0.65} 23.18 _{±0.62}	17.02 _{±0.63} 12.07 _{±1.68}	34.27 _{±1.92} 40.09 _{±1.72}	488.39 _{±18.56} 468.74 _{±19.69}	639.73 _{±28.15} 609.18 _{±28.02}	219.58 _{±5.08} 275.74 _{±10.48}	158.55 _{±51.02} 296.24 _{±96.35}	678.41 _{±64.28} 629.55 _{±59.10}
D5	RMSE 8.88 _{±0.66} MAE 9.24 _{±0.43}	138.95 _{±9.59} 141.5 _{±8.23}	128.5 _{±3.71} 127.79 _{±4.95}	11.71 _{±0.94} 12.66 _{±0.52}	26.98 _{±0.41} 14.68 _{±0.61}	42.2 _{±3.53} 59.06 _{±2.34}	369.61 _{±14.41} 343.85 _{±14.10}	484.94 _{±20.85} 541.95 _{±26.56}	390.7 _{±14.97} 509.75 _{±23.17}	272 _{±16.51} 901.16 _{±11.82}	499.20 _{±58.56} 518.93 _{±62.42}
D6	RMSE 5.33 _{±0.04} MAE 5.33 _{±0.04}	214.96 _{±3.58} 192.14 _{±4.36}	258.28 _{±4.71} 220.03 _{±7.75}	19.57 _{±2.82} 6.34 _{±0.93}	35.77 _{±1.1} 8.82 _{±0.65}	65.38 _{±1.91} 116.52 _{±3.96}	541.70 _{±19.50} 553.36 _{±21.58}	769.56 _{±32.32} 709.00 _{±31.20}	799.05 _{±9.98} 1125.55 _{±5.81}	146.72 _{±26.42} 575.61 _{±93.9}	1162.62 _{±108.12} 1635.87 _{±105.12}
D7	RMSE 4722.89 _{±129.8} MAE 5028.77 _{±425.6}	153965 _{±13366} 171479 _{±23392}	171167 _{±3378} 171614 _{±11662}	7175 _{±311} 7040 _{±233}	22549 _{±398} 14130 _{±926.34}	44765 _{±671} 25803 _{±2207}	371055.65 _{±17439.62} 468137.67 _{±23875.02}	511163.80 _{±27091.68} 624183.56 _{±34954.28}	- -	- -	- -
D8	RMSE 2083.52 _{±110} MAE 3137.18 _{±129}	135316 _{±11325.60} 135923 _{±13284}	139236 _{±2862} 140292 _{±3358}	4646 _{±263} 5592 _{±308}	15175 _{±337} 13190 _{±395}	210273 _{±568} 25836 _{±667}	349115.28 _{±15710.19} 334370.58 _{±15381.05}	474959.16 _{±24697.88} 512429.71 _{±28183.63}	- -	- -	- -
Win/Loss	-	16/0	16/0	16/0	16/0	16/0	16/0	16/0	16/0	16/0	16/0
F-rank	1.00	5.75	5.75	2.25	2.81	4.06	8.44	9.75	8.75	7.50	9.94

TABLE S4
WILCOXON SIGNED-RANK TEST FOR PREDICTION ACCURACY

Comparison	R+	R-	p-value	Rank-biserial correlation
M1 vs M2	136	0	1.53×10^{-5}	1.0
M1 vs M3	136	0	1.53×10^{-5}	1.0
M1 vs M4	136	0	1.53×10^{-5}	1.0
M1 vs M5	136	0	1.53×10^{-5}	1.0
M1 vs M6	136	0	1.53×10^{-5}	1.0
M1 vs M7	136	0	1.53×10^{-5}	1.0
M1 vs M8	136	0	1.53×10^{-5}	1.0
M1 vs M9	136	0	1.53×10^{-5}	1.0
M1 vs M10	136	0	1.53×10^{-5}	1.0
M1 vs M11	136	0	1.53×10^{-5}	1.0

TABLE S5
WILCOXON SIGNED-RANK TEST FOR COMPUTATIONAL EFFICIENCY

Comparison	R+	R-	p-value	Rank-biserial correlation
M1 vs M2	136	0	1.53×10^{-5}	1.0
M1 vs M3	136	0	1.53×10^{-5}	1.0
M1 vs M4	136	0	1.53×10^{-5}	1.0
M1 vs M5	136	0	1.53×10^{-5}	1.0
M1 vs M6	136	0	1.53×10^{-5}	1.0
M1 vs M7	136	0	1.53×10^{-5}	1.0
M1 vs M8	136	0	1.53×10^{-5}	1.0
M1 vs M9	136	0	1.53×10^{-5}	1.0
M1 vs M10	136	0	1.53×10^{-5}	1.0
M1 vs M11	136	0	1.53×10^{-5}	1.0

TABLE S6
THE PERFORMANCE OF FNL ON D1.1–D1.4

Dataset	Metrics	D1.1	D1.2	D1.3	D1.4
M1	RMSE	0.3642 $\pm 9.47E-4$	0.3571 $\pm 8.26E-4$	0.3493 $\pm 7.14E-4$	0.3418 $\pm 5.47E-4$
	*Per	0.012 $\pm 1.13E-3$	0.94 $\pm 4.32E-2$	5.91 ± 0.92	27.84 ± 2.13
	MAE	0.2783 $\pm 3.14E-4$	0.2724 $\pm 2.68E-4$	0.2661 $\pm 2.23E-4$	0.2486 $\pm 1.92E-4$
	**Per	0.01 $\pm 1.22E-3$	0.89 $\pm 5.66E-2$	5.57 ± 0.53	26.26 ± 5.22
M2	RMSE	0.3928 $\pm 1.23E-3$	0.3846 $\pm 1.04E-3$	0.3752 $\pm 9.16E-4$	0.3690 $\pm 8.14E-4$
	*Per	1.25 $\pm 2.15E-2$	25.67 ± 5.77	193.42 ± 43.5	1176.24 ± 376
	MAE	0.2987 $\pm 9.83E-4$	0.2914 $\pm 8.47E-4$	0.2836 $\pm 7.65E-4$	0.2652 $\pm 6.83E-4$
	**Per	1.18 $\pm 1.91E-3$	24.21 ± 4.23	188.09 ± 11.9	1154.64 ± 355
M3	RMSE	0.3714 $\pm 6.23E-4$	0.3637 $\pm 5.14E-4$	0.3561 $\pm 3.96E-4$	0.3482 $\pm 3.47E-4$
	*Per	0.39 $\pm 5.98E-2$	8.92 ± 0.29	139.61 ± 47.4	1594.72 ± 288
	MAE	0.2836 $\pm 5.14E-4$	0.2768 $\pm 4.13E-4$	0.2697 $\pm 2.47E-4$	0.2574 $\pm 2.19E-4$
	**Per	0.37 $\pm 9.43E-2$	8.39 ± 0.65	137.23 ± 12.5	1499.03 ± 250
M4	RMSE	0.3953 $\pm 9.74E-4$	0.3872 $\pm 8.76E-4$	0.3774 $\pm 7.89E-4$	0.3671 $\pm 7.14E-4$
	*Per	0.79 $\pm 4.28E-2$	15.97 ± 1.60	29.47 ± 3.33	42.68 ± 2.43
	MAE	0.3012 $\pm 7.23E-4$	0.2938 $\pm 5.84E-4$	0.2857 $\pm 5.24E-4$	0.2714 $\pm 5.03E-4$
	**Per	0.75 $\pm 7.58E-2$	15.17 ± 1.54	26.49 ± 1.65	30.54 ± 2.31
M5	RMSE	0.3897 $\pm 7.24E-4$	0.3821 $\pm 6.37E-4$	0.3719 $\pm 5.63E-4$	0.3624 $\pm 5.26E-4$
	*Per	0.90 $\pm 3.95E-2$	18.94 ± 1.77	67.38 ± 6.73	172.91 ± 27.3
	MAE	0.2947 $\pm 9.76E-4$	0.2873 $\pm 8.63E-4$	0.2787 $\pm 8.14E-4$	0.2653 $\pm 7.84E-4$
	**Per	0.85 $\pm 5.32E-2$	17.88 ± 1.29	63.63 ± 7.38	157.66 ± 25.8
M6	RMSE	0.4043 $\pm 6.67E-4$	0.3967 $\pm 6.24E-4$	0.3864 $\pm 5.47E-4$	0.3768 $\pm 4.86E-4$
	*Per	0.95 $\pm 9.14E-2$	19.93 ± 1.92	71.26 ± 7.13	288.24 ± 38.8
	MAE	0.3083 $\pm 6.14E-4$	0.3008 $\pm 5.27E-4$	0.2923 $\pm 4.92E-4$	0.2787 $\pm 4.67E-4$
	**Per	0.90 $\pm 5.96E-3$	18.88 ± 1.19	67.29 ± 6.73	272.15 ± 92.2
M7	RMSE	0.3689 $\pm 8.92E-4$	0.3628 $\pm 7.94E-4$	0.3556 $\pm 6.82E-4$	0.3466 $\pm 5.91E-4$
	*Per	3.02 $\pm 3.86E-2$	67.00 ± 4.91	452.60 ± 36.7	3269.95 ± 186
	MAE	0.2816 $\pm 3.66E-4$	0.2765 $\pm 3.14E-4$	0.2706 $\pm 2.75E-4$	0.2518 $\pm 2.28E-4$
	**Per	2.81 $\pm 3.42E-2$	62.22 ± 4.36	462.70 ± 31.2	3152.17 ± 169
M8	RMSE	0.3718 $\pm 9.21E-4$	0.3657 $\pm 8.31E-4$	0.3587 $\pm 7.35E-4$	0.3493 $\pm 6.13E-4$
	*Per	4.20 $\pm 4.93E-2$	95.49 ± 6.38	673.10 ± 51.8	4599.10 ± 264
	MAE	0.2839 $\pm 3.94E-4$	0.2787 $\pm 3.45E-4$	0.2730 $\pm 3.04E-4$	0.2538 $\pm 2.53E-4$
	**Per	3.91 $\pm 4.27E-2$	88.12 ± 5.97	667.72 ± 47.6	4456.91 ± 241
M9	RMSE	0.6276 $\pm 8.47E-4$	0.6189 $\pm 7.86E-4$	0.6097 $\pm 7.26E-4$	0.5983 $\pm 6.67E-4$
	*Per	2.15 $\pm 1.98E-3$	34.67 ± 0.67	120.92 ± 21.42	1592.7 ± 159
	MAE	0.4554 $\pm 9.47E-4$	0.4472 $\pm 8.74E-4$	0.4391 $\pm 8.24E-4$	0.4251 $\pm 7.93E-4$
	**Per	2.14 $\pm 1.56E-3$	34.39 ± 0.39	119.67 ± 64.29	1497.1 ± 150
M10	RMSE	0.6221 $\pm 5.73E-4$	0.6139 $\pm 5.24E-4$	0.6043 $\pm 4.83E-4$	0.5927 $\pm 4.36E-4$
	*Per	2.16 $\pm 1.67E-3$	35.32 ± 5.37	186.93 ± 18.7	1783.2 ± 179
	MAE	0.4513 $\pm 3.67E-4$	0.4417 $\pm 3.34E-4$	0.4324 $\pm 3.04E-4$	0.4212 $\pm 2.67E-4$
	**Per	2.15 $\pm 1.57E-3$	35.01 ± 5.79	175.72 ± 17.23	1676.2 ± 111
M11	RMSE	0.4852 $\pm 6.67E-4$	0.4760 $\pm 6.24E-4$	0.4637 $\pm 5.47E-4$	0.4522 $\pm 4.86E-4$
	*Per	2.365 $\pm 9.14E-2$	38.14 ± 1.92	133.01 ± 7.13	1751.97 ± 38.8
	MAE	0.3699 $\pm 6.14E-4$	0.3609 $\pm 5.27E-4$	0.3507 $\pm 4.92E-4$	0.3344 $\pm 4.67E-4$
	**Per	2.354 $\pm 5.96E-3$	37.83 ± 1.19	131.64 ± 6.73	1647.81 ± 92.2

*/**Time cost per training in RMSE/MAE (Seconds).

TABLE S7
PERFORMANCE OF FNL WITH MANUALLY TUNED AND SELF-ADAPTED HYPERPARAMETERS.

Datasets		Manual-tuning	Self-adaptation	Gap
D1	RMSE	$0.3390 \pm 4.32E-04$	0.3387 $\pm 6.80E-04$	0.08%
	*Per	5802.11 ± 48.73	37.48 ± 0.66	99.35%
	MAE	$0.2452 \pm 2.77E-04$	0.2450 $\pm 1.81E-04$	0.08%
	**Per	8123.96 ± 29.15	39.82 ± 0.55	99.50%
D2	RMSE	$0.3469 \pm 5.29E-05$	0.3466 $\pm 1.61E-05$	0.08%
	*Per	2613.86 ± 25.76	22.74 ± 0.35	99.12%
	MAE	$0.2477 \pm 3.64E-05$	0.2474 $\pm 1.36E-05$	0.12%
	**Per	3267.33 ± 17.63	24.17 ± 2.35	99.26%
D3	RMSE	0.3189 $\pm 7.19E-05$	$0.3192 \pm 3.93E-06$	0.09%
	*Per	1058.04 ± 34.17	4.99 ± 0.43	99.52%
	MAE	0.2184 $\pm 3.80E-06$	$0.2187 \pm 2.12E-06$	0.09%
	**Per	1143.65 ± 56.56	3.23 ± 0.32	99.71%
D4	RMSE	$0.3164 \pm 4.51E-06$	0.3161 $\pm 6.49E-06$	0.09%
	*Per	719.59 ± 9.77	8.03 ± 0.55	98.88%
	MAE	$0.2167 \pm 3.33E-06$	0.2166 $\pm 2.78E-06$	0.09%
	**Per	770.99 ± 5.61	4.59 ± 0.66	99.41%
D5	RMSE	0.2900 $\pm 5.29E-04$	$0.2904 \pm 2.18E-04$	0.13%
	*Per	1010.39 ± 11.8	8.88 ± 0.66	99.12%
	MAE	$0.1998 \pm 1.50E-05$	0.1996 $\pm 1.81E-04$	0.10%
	**Per	1309.77 ± 18.74	9.24 ± 0.43	99.29%
D6	RMSE	$0.3430 \pm 2.14E-04$	0.3426 $\pm 5.61E-06$	0.14%
	*Per	2280.92 ± 32.15	5.33 ± 0.04	99.76%
	MAE	0.2488 $\pm 5.37E-04$	$0.2493 \pm 2.98E-06$	0.16%
	**Per	2348.05 ± 18.99	5.33 ± 0.04	99.77%
D7	RMSE	$0.3250 \pm 3.21E-04$	0.3245 $\pm 2.22E-04$	0.15%
	*Per	590361 ± 3761	4722.89 ± 129.8	99.20%
	MAE	$0.2298 \pm 5.01E-04$	0.2294 $\pm 3.85E-04$	0.17%
	**Per	678780 ± 1120	5028.77 ± 425.6	99.26%
D8	RMSE	$0.3199 \pm 5.01E-04$	0.3117 $\pm 4.00E-04$	0.06%
	*Per	121296 ± 407	1083.52 ± 110	99.10%
	MAE	0.2162 $\pm 3.10E-04$	$0.2164 \pm 2.98E-04$	0.09%
	**Per	140988 ± 356	1137.18 ± 129	99.19%

*/**Time cost per training in RMSE/MAE (Seconds).

TABLE S8
PERFORMANCE OF FNL UNDER DIFFERENT FUZZY-CONTROLLER DESIGN CHOICES ON D1 AND D8.

Design factor	Setting	D1				D8			
		RMSE	Time (s)	MAE	Time (s)	RMSE	Time (s)	MAE	Time (s)
Number of fuzzy levels	3	0.3452	29.36	0.2513	31.25	0.3189	1652.41	0.2228	2479.63
	4	0.3414	33.81	0.2484	35.74	0.3143	1847.65	0.2187	2786.42
	6	0.3387	37.48	0.2450	39.82	0.3117	2083.52	0.2164	3137.18
	8	0.3391	44.67	0.2455	47.12	0.3120	2446.31	0.2167	3671.54
Balance coefficient	0.25	0.3401	36.82	0.2464	39.15	0.3130	2027.60	0.2175	3098.40
	0.50	0.3387	37.48	0.2450	39.82	0.3117	2083.52	0.2164	3137.18
	0.75	0.3395	39.51	0.2459	41.73	0.3125	2184.30	0.2170	3276.80
Membership function	Triangular	0.3387	37.48	0.2450	39.82	0.3117	2083.52	0.2164	3137.18
	Gaussian	0.3393	42.16	0.2457	44.83	0.3123	2297.84	0.2169	3448.56

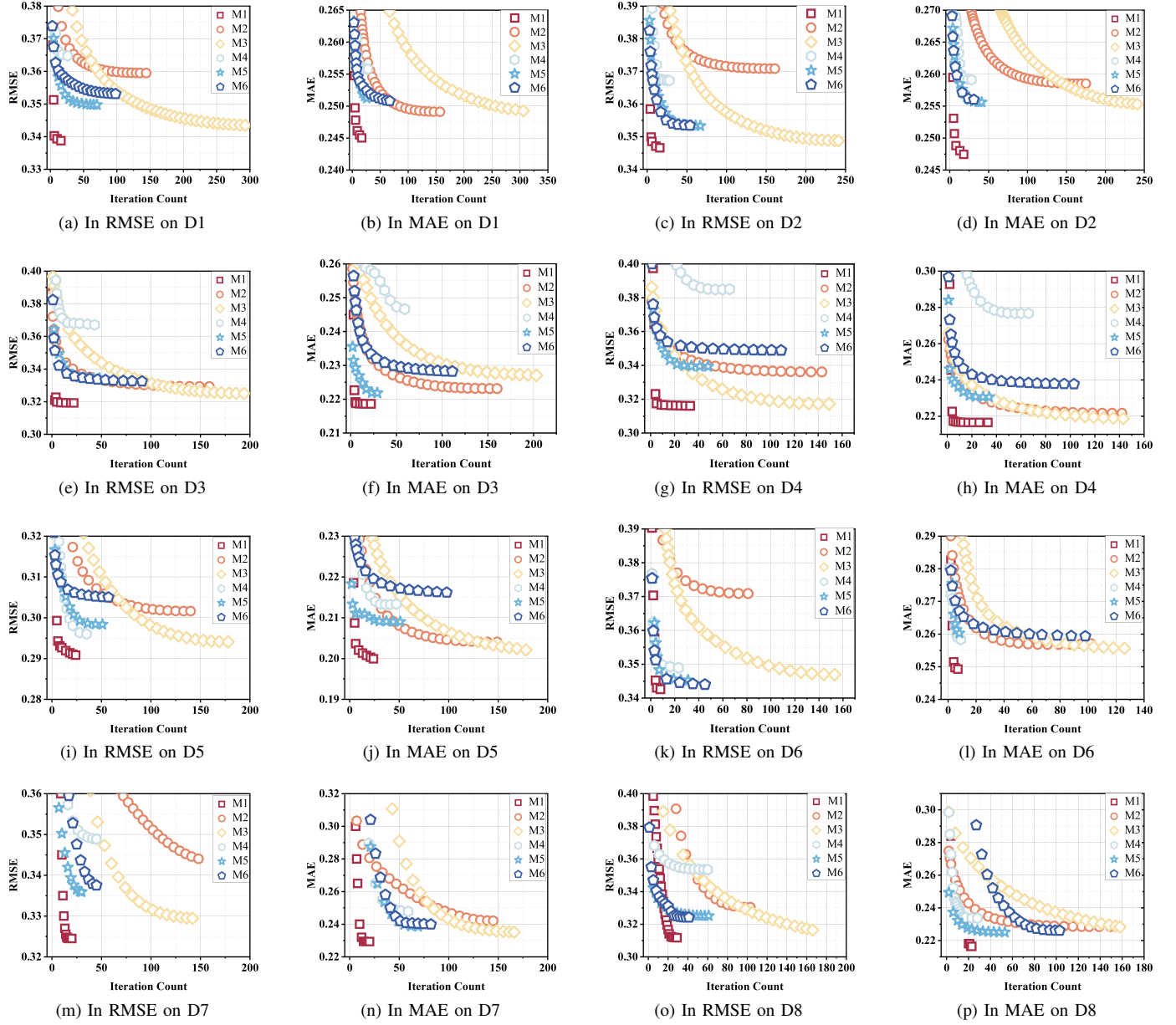


Fig. S1. Training curves of M1-M6 for RMSE and MAE on D1-D8.

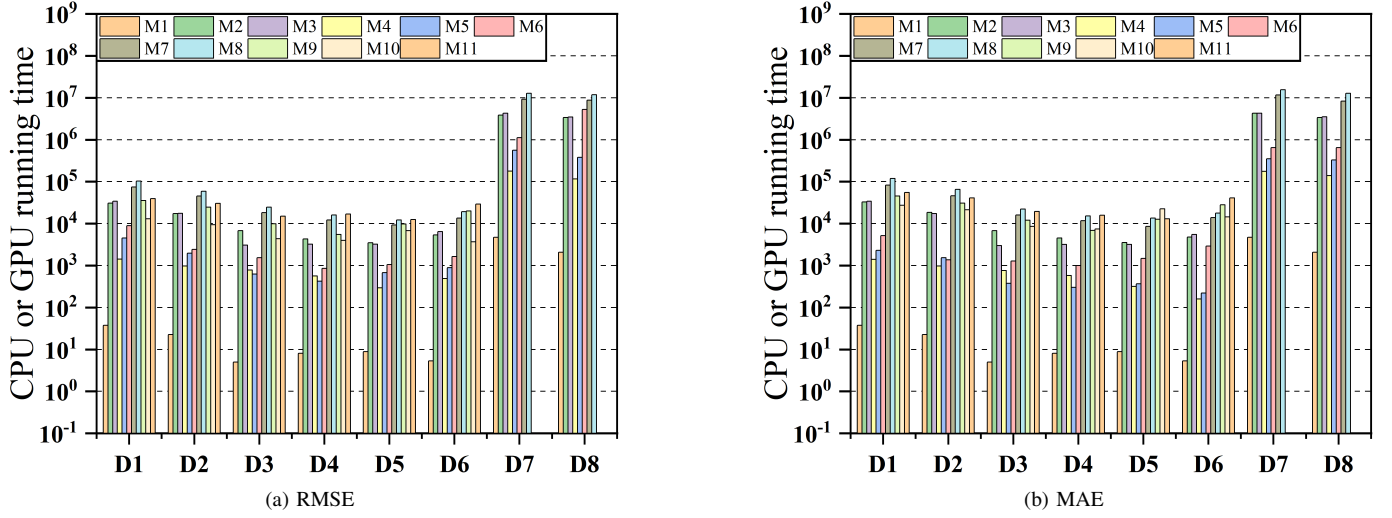
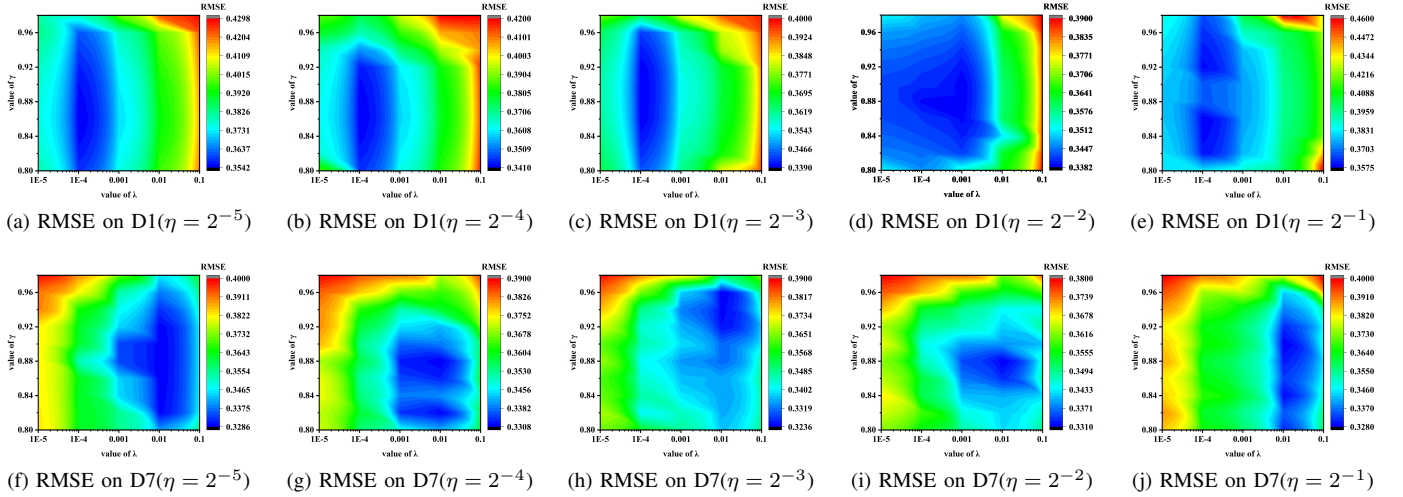
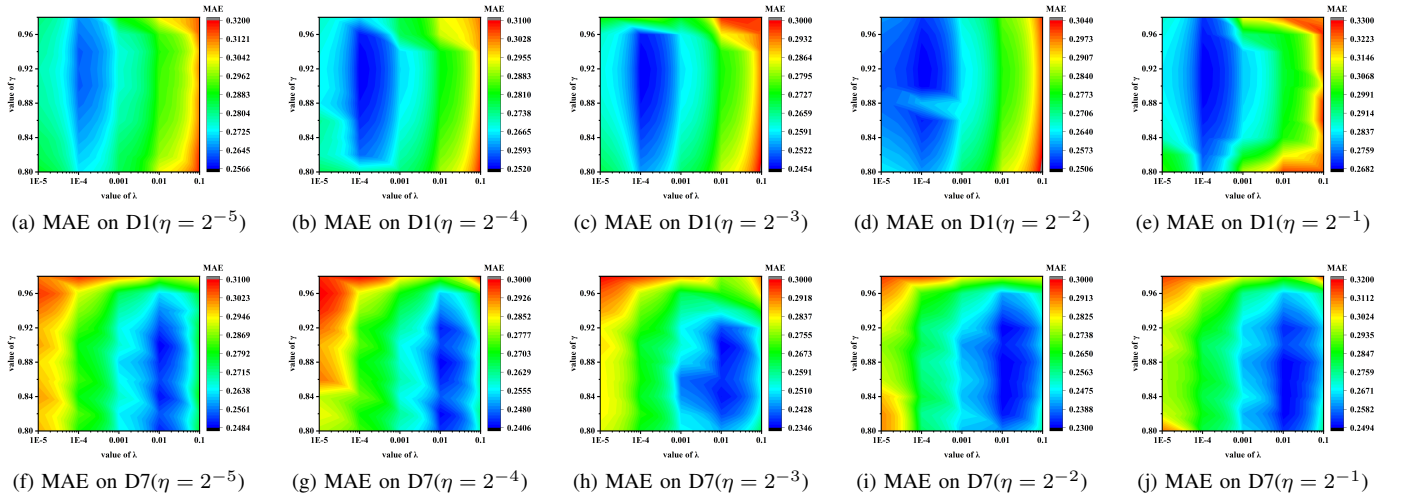


Fig. S2. Total time cost of M1–M11 for obtaining the lowest RMSE and MAE on D1–D8.

Fig. S3. RMSE of FNL as η , λ and γ vary on D1 and D7.Fig. S4. MAE of FNL as η , λ and γ vary on D1 and D7.

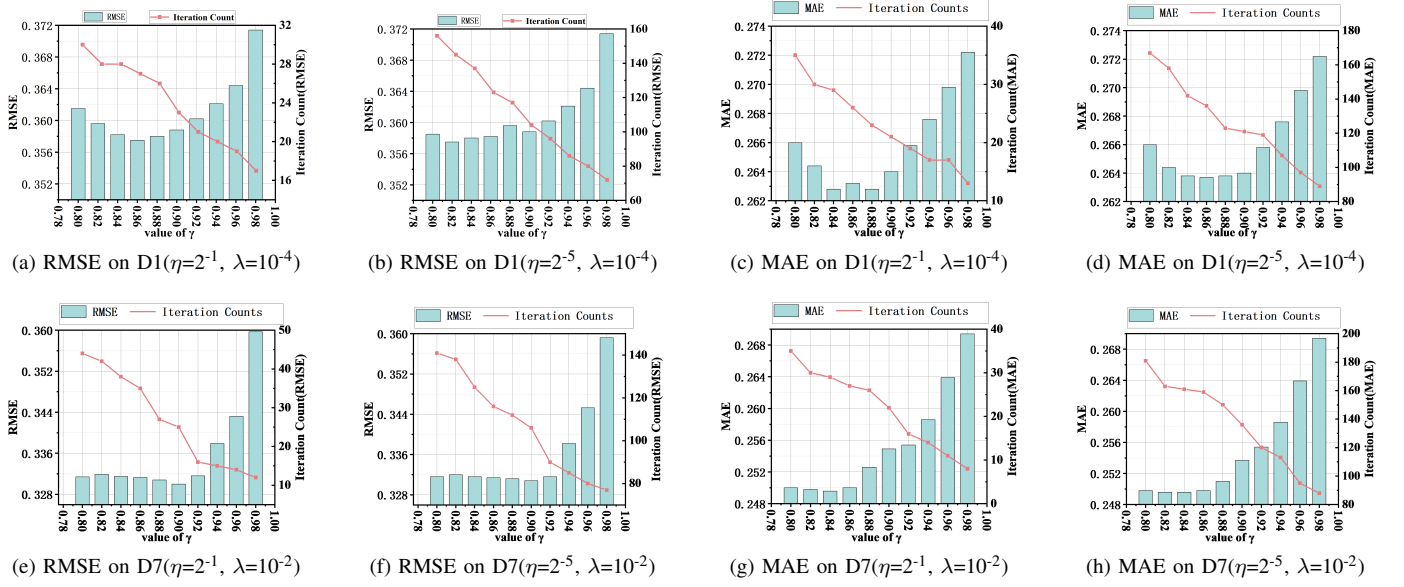
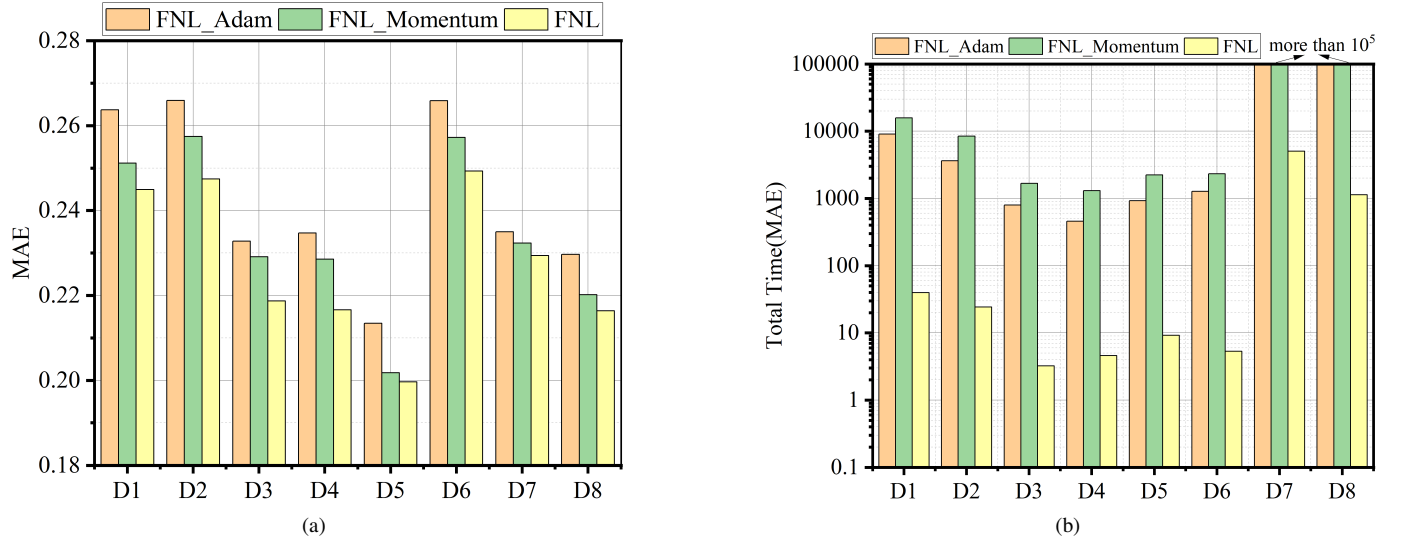
Fig. S5. RMSE/MAE and iterations of FNL as η , λ and γ vary on D1 and D7.

Fig. S6. MAE and corresponding total time cost of FNL with different optimization methods on D1–D8.